Latest updates: https://dl.acm.org/doi/10.1145/3674399.3674453

RESEARCH-ARTICLE

Diagnosis of Alzheimer's Disease Based on Structural Graph Convolutional Neural Networks

HUAN LAO, Guangxi University for Nationalities, Nanning, Guangxi, China

HONGFEI JIA

ZHENHAI CHEN, Guangxi University for Nationalities, Nanning, Guangxi, China

Open Access Support provided by:

Guangxi University for Nationalities

PDF Download
3674399.3674453.pdf
15 February 2026
Total Citations: 1
Total Downloads: 427

Published: 05 July 2024

Citation in BibTeX format

ACM-TURC '24: ACM Turing Award Celebration Conference 2024
July 5 - 7, 2024
Changsha, China

ACM-TURC '24: Proceedings of the ACM Turing Award Celebration Conference - China 2024 (July 2024)
https://doi.org/10.1145/3674399.3674453
ISBN: 9798400710117

Diagnosis of Alzheimer's Disease Based on Structural Graph Convolutional Neural Networks

Huan Lao*
laohuanlh@gxmzu.edu.cn
School of Artificial Intelligence,
Guangxi Minzu University
Guangxi Key Lab of Multi-source
Information Mining & Security
Nanning, Guangxi, China

Hongfei Jia
Beijing ZhongkeHaixun Digital
Technology Co.,Ltd
Beijing, China

Zhenhai Chen
School of Artificial Intelligence,
Guangxi Minzu University
Nanning, Guangxi, China

ABSTRACT

In recent years, classification methods based on multi-modality images have been widely applied in the diagnosis of Alzheimer's Disease(AD), and have achieved better performance than methods based on single-modality image. However, most multi-modality based classification methods only extract features from a single-modality image independently, and then concatenate them into a long vector for classification, without properly considering the correlation between different modalities. Based on this, this study proposes a structure graph convolutional neural network(GCN) method for AD classification. Firstly, using 90 regions of interest(ROIs) from different modal images as nodes in the graph, extract the positional relationships and specific features of ROIs to construct a structural diagram. Then, the GCN network is used to aggregate the adjacent node features of the structural graph, and pooling layer is used to obtain graph representations representing different modal image features. Finally, the final AD classification diagnosis result is output through the softmax layer. Experiments results show that this method has better classification performance than existing methods.

CCS CONCEPTS

• **Computing methodologies → Image processing**; • **Applied computing → Imaging**; **Bioinformatics**.

KEYWORDS

Alzheimer's disease, graph convolutional neural networks, multi-modality classification

ACM Reference Format:
Huan Lao, Hongfei Jia, and Zhenhai Chen. 2024. Diagnosis of Alzheimer's Disease Based on Structural Graph Convolutional Neural Networks. In *ACM Turing Award Celebration Conference 2024 (ACM-TURC '24), July 05–07, 2024, Changsha, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3674399.3674453

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
ACM-TURC '24, July 05–07, 2024, Changsha, China
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1011-7/24/07
https://doi.org/10.1145/3674399.3674453

1 INTRODUCTION

Alzheimer's disease (AD) is a common neurodegenerative disease in the elderly, which patients gradually lose their memory, language, and action abilities as the condition progresses[6]. With the continuous development of medical imaging technology, brain images such as structural magnetic resonance imaging (sMRI), diffusion tensor imaging (DTI), functional magnetic resonance imaging (fMRI), and positron emission computed tomography (PET) have been widely used by clinical doctors for the clinical diagnosis of AD[5].

With the development of machine learning, combining brain image and machine learning for AD classification has become a research hotspot. However, most AD classification studies based on brain image and machine learning only use single-modality brain image for research. For example, Lian et al.[7] proposed a hierarchical fully convolutional network (H-FCN) to automatically recognize region of interest(ROI) positions in sMRI images for constructing AD classification. Zhang et al. [13, 14] used sMRI images and Support Vector Machines (SVM) for AD classification. While Illán et al. [4] and Padilla et al.[10] only used FDG-PET images and SVM for AD classification. In recent years, an increasing number of studies have shown that different modal images can provide supplementary information [1, 9]. Among them, sMRI images can provide structural information of the patient's brain, and FDG-PET images can provide metabolic information of the patient's brain. Therefore, using sMRI images and FDG-PET images simultaneously can achieve complementary information and improve the accuracy of AD diagnosis. For example, Liu et al. [8], Huang et al. [3] and Pan et al. [11] combined sMRI images and FDG-PET images to classify AD, and achieved classification accuracy of 93.10%, 90.10% and 92.50%, respectively.

However, how to effectively integrate these multi-modality data in multi-modality AD classification methods to understand the pathology of AD remains a challenge. Many early studies simply combined the feature of different modal data, which was clearly insufficient to uncover hidden information related to pathology. In response to this issue, this study proposes a graph convolutional neural network (GCN) based method for AD classification, in order to better capture the correlation between sMRI images and FDG-PET images. Specifically, firstly, 90 ROIs of different modal images are used as nodes in the graph, the actual position contact relationship of ROIs is used as the adjacency matrix of the graph, and the extracted features are used as node attributes to construct the structural graph. Then, a GCN network is used to aggregate the adjacent node features of the structural graph, and a pooling layer

is used to obtain the graph representation representing the features of different modal images. Finally, a softmax layer is used to output the final AD classification diagnosis result. The experiments on the alzheimer's disease neuroimaging initiative (ADNI) dataset show that the proposed method can effectively capture the relevant information between different modal data, which helps to improve the accuracy of AD/CN classification. The remaining parts of this article are organized as follows. In section 2, we provided a detailed introduction to the dataset and data preprocessing methods used for the study. In section 3, we provide a detailed description of the overall framework of this study. In section 4, our proposed method was evaluated and compared with existing methods. Finally, it is summarized in section 5.

2 MATERIALS

The multi-modality image data used in this study was from the ADNI database (adni.loni.usc.edu). ADNI is a global study that explores the pathogenesis of AD and develops treatment methods that delay or prevent its progression. As of 2023, the ADNI dataset has undergone four stages of research, namely ADNI-1, ADNI-GO, ADNI-2, and ADNI-3. This study selected 325 subjects with both baseline sMRI images and baseline FDG PET images from the ADNI-2 stage, including 144 AD subjects and 181 CN subjects. The formats of the sMRI images and FDG PET images were "MT1; GradWarp; N3m" and "Coreg, Avg, Std Img and Vox Siz, Uniform Resolution", respectively. For all sMRI images, we used the "Segment Data" module in the Computation Anatomy Toolbox (CAT12) toolkit to perform skull stripping, tissue segmentation, and alignment and modulation operations, and obtained a gray matter image (GM-MRI) of size 121×145×121. For all FDG PET images, we used the "FLIRT" module in FMRIB Software Library (FSL) software to register the FDG PET images of each subject onto their respective preprocessed sMRI images. The demographic information for all subjects is shown in Table 1.

3 METHOD

This study focuses on different modal images (3D sMRI images and 3D FDG PET images) and constructs a GCN network to classify AD and CN. The classification framework is shown in Figure 1.

3.1 Three-Dimensional Discrete Wavelet Transform

The three-dimensional discrete wavelet transform (3D-DWT) is an extension of 2D-DWT, which considers the correlation between three-dimensional data [2]. Therefore, performing 3D-DWT decomposition on 3D images helps to understand the detailed information of different directions in 3D images. The schematic diagram of using one-level 3D-DWT to decompose 3D images is shown in Figure 2. Specifically, each 3D image volume is firstly filtered along the x-dimension, resulting in a low-pass image L and a high-pass image H. Both L and H are then filtered along the y-dimension, resulting in four decomposed subbands (LL, LH, HL and HH). Then each of these four subbands are filtered along the z-dimension, resulting in eight subbands (LLL, LLH, LHL, LHH, HLL, HLH, HHL and HHH). Among them, LLL is an approximation of the 3D image and usually contains more information about the image. The other seven subbands are detail components that only contain the detailed information of the image. In this study, we divided the GM-MRI images and FDG PET images of each subject into 90 ROIs using the automated anatomic labeling (AAL) brain atlas. Then, apply the three-level 3D-DWT to each ROI and use the decomposed LLL subband of each ROI as the final feature representation. The two-level 3D-DWT is achieved by decomposing the approximate component of the one-level 3D-DWT, while the three-level 3D-DWT is achieved by decomposing the approximate component of the two-level 3D-DWT.

3.2 Construction of Structural Graph

A graph is a type of non-euclidean space data that consists of nodes and edges, including node information and relationship information between different nodes. Graphs are ubiquitous in the real world, such as social networks, citation networks and biological networks. They usually use a single sample as a node and the relationship between samples as a connection to build a graph. In this study, when constructing a graph, ROIs are used as the node of the graph. When ROIs have actual positional contact relationships, their corresponding nodes are connected to form the edges of the graph. Suppose that X_1 and X_2 represent the set of non-zero pixel position values of the two ROIs respectively, then the actual position contact relation $A_{1,2}$ between the two ROIs can be expressed by finding the intersection between X_1 and X_2, the formula is as follows:

$$A_{1,2} = \begin{cases} 0, & X_1 \cap X_2 = \emptyset \\ 1, & X_1 \cap X_2 \neq \emptyset \end{cases} \tag{1}$$

According to formula (1), create a matrix based on the actual positional contact relationship between ROIs to reflect the correlation between different ROIs, and use it as the adjacency matrix A of the graph (as shown in Figure 3). Then, the 3D-DWT features of different modal images are used as node features of the graph. Finally, the structural graph of each subject is constructed. Each subject's structural graph will have the same fixed graph structure (defined by the adjacency matrix), with the difference being the node values (represented by the extracted 3D-DWT features of different modal images). The structural graph of each subject constructed in this study is shown in Figure 4.

3.3 Graph Convolutional Neural Network

With the increasing number of graph data in daily life, the translation invariance of convolutional neural network (CNN) is not applicable to graphs. GCN network is a deep learning method proposed for graphs, which is a derivative model of Graph Neural Networks (GNN) and CNN. Its contribution is to extend convolutional operations from euclidian space to non-euclidian space. The core idea of GCN is to aggregate and update node features by utilizing the information of edges in the graph, in order to obtain new and higher-level node feature representations. The GCN architecture used in this study consists of graph convolutional layer, average pooling layer, and softmax layer (as shown in Figure 1). Each convolutional layer followed by ReLU activation to in crease non-linearity, the average pooling layer obtains potential representations, and the final classification is achieved through the softmax layer. In this study, the asymmetric edge function EdgeConv[12]

Table 1: Demographic information of the subjects in this study

	Gender(M/F)	Age	Education	MMSE	ADAS-Cog 11	ADAS-Cog 13
AD	83/61	74.56±8.16	15.75±2.66	23.06±2.10	20.84±7.18	31.15±8.45
CN	87/94	73.50±6.27	16.59±2.53	29.04±1.22	5.80 ±3.07	9.09 ±4.51

Note. Data are presented as mean and standard deviation; MMSE=Mini-Mental State Examination; ADAS-Cog=Alzheimer's Disease Assessment Scale-Cognitive.

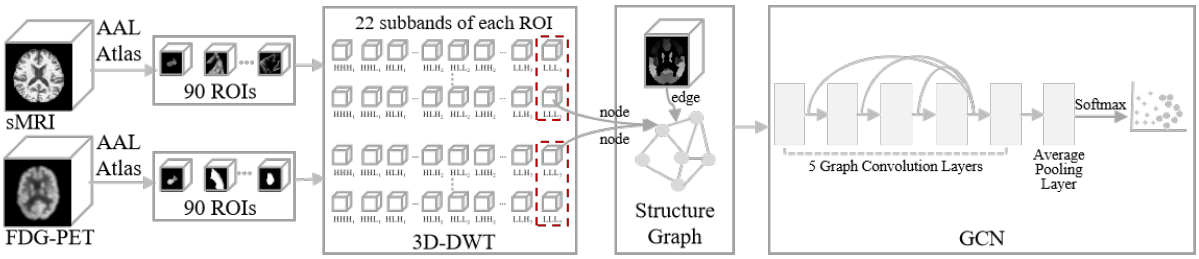

Figure 1: Classification Framework Diagram.

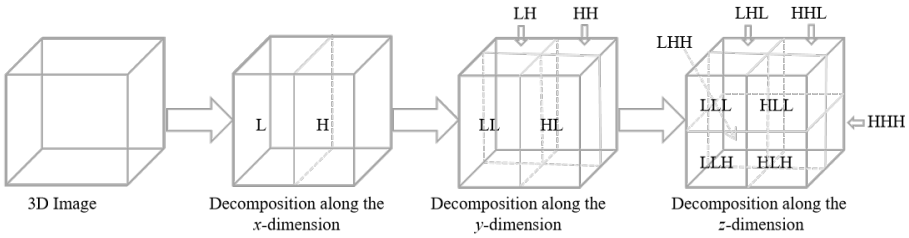

Figure 2: Diagram of one-level 3D-DWT decomposition.

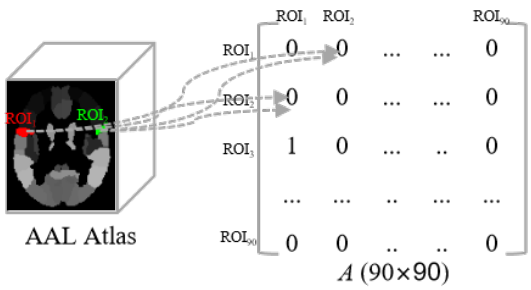

Figure 3: Schematic diagram of adjacency matrix.

is chosen as the convolution operation of the graph convolution layer:

$$x_i^{'} = \max h_{\Theta}(x_i || x_j - x_i) \tag{2}$$

Where x_i is the feature of node i, and x_j is the feature of the connected adjacent node j. In addition to the original features from the nodes, the difference between the nodes $(x_j - x_i)$ is also appended as complementary features. h_θ represents a trainable weight θ with a nonlinear activation function h. This function is implemented by a multi-layer perceptron (MLP) with a corrected linear unit (ReLU).

4 EXPERIMENTS

To validate the effectiveness of our method, we conducted AD vs. CN classification experiments on the ADNI-2 dataset. The experiment used 10-fold cross validation to avoid the impact of bias caused by randomly dividing the dataset on classification performance. The classification performance was evaluated using classification accuracy (ACC), sensitivity (SEN), specificity (SPE), and area under the receiver operating characteristic curve (AUC). The experimental results are shown in Table 2. As can be observed from Table 2, The classification performance of multi-modality images (ACC of 93.12%, SEN of 94.85%, SPE of 90.42%, AUC of 97.08%) is better than that of single-modality images (ACC of 90.56%/90.87%, SEN of 89.58%/92.55%, SPE of 91.75%/88.26%, AUC of 96.13%/94.97%). This is because the multi-modality image features realize the complementary information and provide more effective information.

In addition, in Table 2, we compared the classification results of the proposed method with the existing AD classification methods based on single-modality image[4, 7, 10, 13, 14] and multi-modality images[3, 8, 11]. In order to ensure the optimal classification performance for each method, the classification results for each method in the table are the best results obtained by experiments on its original selected subset of data. That is to say, the number of subjects and the division of training and testing samples are different between these methods. Although the selection of images by all comparison methods is not entirely the same, they were all studied

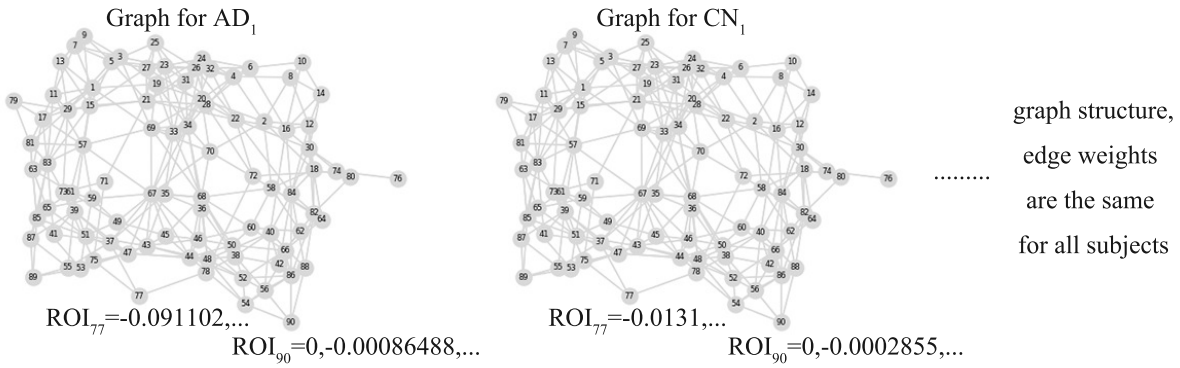

Figure 4: Schematic diagram of graph structure for each subject.

Table 2: Comparison of the classification accuracy of our method with the existing methods

Method	Images	AD vs. CN Classification Results(%)			
		ACC	SEN	SPE	AUC
Lian et al.[7]	sMRI	90.3	82.4	96.5	95.1
Zhang et al.[13]	sMRI	83.7	80.9	86.7	-
Zhang et al.[14]	sMRI	88.3	79.61	94.69	94.01
Proposed method	sMRI	**90.56**	**89.58**	91.75	**96.13**
Illán et al.[4]	FDG PET	88.24	88.64	87.7	-
Padilla et al.[10]	FDG PET	86.59	87.5	85.36	-
Proposed method	FDG PET	**90.87**	**92.55**	**88.26**	**94.97**
Liu et al.[8]	sMRI+FDG PET	93.1	90	95.65	94.83
Huang et al.[3]	sMRI+FDG PET	90.1	90.85	89.21	90.84
Pan et al.[11]	sMRI+FDG PET	92.5	89.94	94.53	95.89
Proposed method	sMRI+FDG PET	**93.12**	**94.85**	90.42	**97.08**

using the ADNI dataset, and the ADNI research group conducted quality control and preprocessing on the images. It can be observed from Table 2 that our proposed method outperforms some existing methods in AD and CN classification. Specifically, the classification accuracy of our proposed method is greater than 90% with both single-modality image and multi-modality images. Compared with the classification method using sMRI images, our proposed method achieves a classification accuracy of 90.56%, which is 0.26%, 6.86%, and 2.26% higher than existing methods[7, 13, 14] . Compared with the classification method using FDG PET images[4, 10] , the classification accuracy of our proposed method is improved by 2.63% and 4.28%. Compared with the classification method using multi-modality images[3, 8, 11], the classification accuracy of our proposed method is improved by 3.02%, 0.02% and 0.62%.

5 CONCLUSION

In this study, we propose a structure graph convolutional neural network method for classifying AD and CN, including data preprocessing, 3D-DWT feature extraction, structure graph construction, and graph convolutional neural network classification. To validate the effectiveness of the proposed method, we conducted classification experiments using baseline sMRI images and baseline FDG PET images from 325 subjects on the ADNI-2 dataset. The experimental results show that the proposed method can jointly extract common features of multi-modality brain images and has better classification performance than existing methods. However, our method still has certain limitations. For example, the ADNI dataset collected demographic, neuropsychological, imaging, genetic, cerebrospinal fluid, and blood data from various subjects according to

unified standards, while our method only used imaging data for classification. In the future, we will combine more modal data to classify AD.

ACKNOWLEDGMENTS

The data used in preparation of this study was obtained from the Alzheimer's Disease Neuroimaging Initiative(ADNI). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data, but did not participate in the analysis or writing of this report.

This work was supported by the Guangxi Minzu University Scientific Research Fund Grant Project (Scientific Research Initiation Project for Introduced Talents & 2022KJQD21), the Research Fund of Guangxi Key Lab of Multi-source Information Mining & Security (MIMS22-09), and the Project of Improving the Basic Scientific Research Ability of Young and Middle-Aged Teachers in Guangxi Universities(2023KY0170).

REFERENCES

[1] Yanjiao Ban, Huan Lao, Bin Li, Wenjun Su, and Xuejun Zhang. 2023. Diagnosis of Alzheimer's disease using hypergraph p-Laplacian regularized multi-task feature learning. *Journal of Biomedical Informatics* 140 (2023), 104326. https://doi.org/10.1016/j.jbi.2023.104326

[2] Aman Ghasemzadeh and Hasan Demirel. 2018. 3D discrete wavelet transform-based feature extraction for hyperspectral face recognition. *IET Biometrics* 7, 1 (2018), 49–55. https://doi.org/10.1049/iet-bmt.2017.0082

[3] Yechong Huang, Jiahang Xu, Yuncheng Zhou, Tong Tong, Xiahai Zhuang, and the Alzheimer's Disease Neuroimaging Initiative (ADNI) . 2019. Diagnosis of Alzheimer's Disease via Multi-Modality 3D Convolutional Neural Network. *Frontiers in Neuroscience* 13 (2019), 1–12. https://doi.org/10.3389/fnins.2019.00509

[4] I.A. Illán, J.M. Górriz, J. Ramírez, D. Salas-Gonzalez, M.M. López, F. Segovia, R. Chaves, M. Gómez-Rio, and C.G. Puntonet. 2011. 18F-FDG PET imaging analysis for computer aided Alzheimer's diagnosis. *Information Sciences* 181, 4 (2011), 903–916. https://doi.org/10.1016/j.ins.2010.10.027

[5] Huan Lao and Xuejun Zhang. 2022. Diagnose Alzheimer's disease by combining 3D discrete wavelet transform and 3D moment invariants. *IET Image Processing* 16, 14 (2022), 3948–3964. https://doi.org/10.1049/ipr2.12605

[6] Huan Lao and Xuejun Zhang. 2022. Regression and Classification of Alzheimer's Disease Diagnosis Using NMF-TDNet Features From 3D Brain MR Image. *IEEE Journal of Biomedical and Health Informatics* 26, 3 (2022), 1103–1115. https://doi.org/10.1109/JBHI.2021.3113668

[7] Chunfeng Lian, Mingxia Liu, Jun Zhang, and Dinggang Shen. 2020. Hierarchical Fully Convolutional Network for Joint Atrophy Localization and Alzheimer's Disease Diagnosis Using Structural MRI. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 4 (2020), 880–893. https://doi.org/10.1109/TPAMI.2018.2889096

[8] Mingxia Liu, Jun Zhang, Pew-Thian Yap, and Dinggang Shen. 2017. View-aligned hypergraph learning for Alzheimer's disease diagnosis with incomplete multi-modality data. *Medical Image Analysis* 36 (2017), 123–134. https://doi.org/10.1016/j.media.2016.11.002

[9] Siqi Liu, Sidong Liu, Weidong Cai, Hangyu Che, Sonia Pujol, Ron Kikinis, Dagan Feng, Michael J. Fulham, and ADNI. 2015. Multimodal Neuroimaging Feature Learning for Multiclass Diagnosis of Alzheimer's Disease. *IEEE Transactions on Biomedical Engineering* 62, 4 (2015), 1132–1140. https://doi.org/10.1109/TBME.2014.2372011

[10] P. Padilla, M. Lopez, J. M. Gorriz, J. Ramirez, D. Salas-Gonzalez, and I. Alvarez. 2012. NMF-SVM Based CAD Tool Applied to Functional Brain Images for the Diagnosis of Alzheimer's Disease. *IEEE Transactions on Medical Imaging* 31, 2 (2012), 207–216. https://doi.org/10.1109/TMI.2011.2167628

[11] Yongsheng Pan, Mingxia Liu, Chunfeng Lian, Tao Zhou, Yong Xia, and Dinggang Shen. 2018. Synthesizing Missing PET from MRI with Cycle-consistent Generative Adversarial Networks for Alzheimer's Disease Diagnosis. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Alejandro F. Frangi, Julia A. Schnabel, Christos Davatzikos, Carlos Alberola-López, and Gabor Fichtinger (Eds.). Springer International Publishing, Cham, 455–463.

[12] Yue Wang, Yongbin Sun, Ziwei Liu, Sanjay E. Sarma, Michael M. Bronstein, and Justin M. Solomon. 2019. Dynamic Graph CNN for Learning on Point Clouds. *ACM Trans. Graph.* 38, 5, Article 146 (oct 2019), 12 pages. https://doi.org/10.1145/3326362

[13] Jun Zhang, Yue Gao, Yaozong Gao, Brent C. Munsell, and Dinggang Shen. 2016. Detecting Anatomical Landmarks for Fast Alzheimer's Disease Diagnosis. *IEEE Transactions on Medical Imaging* 35, 12 (2016), 2524–2533. https://doi.org/10.1109/TMI.2016.2582386

[14] Jun Zhang, Mingxia Liu, Le An, Yaozong Gao, and Dinggang Shen. 2017. Alzheimer's Disease Diagnosis Using Landmark-Based Features From Longitudinal Structural MR Images. *IEEE Journal of Biomedical and Health Informatics* 21, 6 (2017), 1607–1616. https://doi.org/10.1109/JBHI.2017.2704614